# A CNN-Transformer Embedded Unfolding Network for Hyperspectral Image Super-Resolution

Yao Tang, Jie Li, *Member, IEEE*, Linwei Yue, *Member, IEEE*, Xinxin Liu, *Member, IEEE*, Yajie Li, Yi Xiao, *Graduate Student Member, IEEE*, and Qiangqiang Yuan, *Member, IEEE*

***Abstract*— Hyperspectral images (HSIs) with rich spectral information have been widely used in surface classification, object detection, and other real application problems. However, due to the hardware limitations, the low spatial resolution HSIs hinder the exploration of their application potential. Deep learning-based methods are currently the most common solutions for single HSI super-resolution (HSI SR) tasks. However, such methods often overlook the degradation principle from high-resolution HSI to low-resolution HSI. In this article, we propose a CNN-transformer embedded unfolding network (CTUNet), in which an unfolding framework with an effective spatial–spectral prior network is designed for HSI SR by incorporating the degradation principle of HSIs. Specifically, a maximum posterior-based energy model is employed, enabling alternate optimization to seek the optimal solution in an iterative mechanism. To effectively utilize the structure prior of HSI, multiscale self-calibrated convolution (MSSC) and edge-guided transformer module are combined to learn latent spatial–spectral priors. Additionally, hidden feature connections between adjacent iterations enhance the representation of the image features. Extensive experiments conducted on three available HSI datasets demonstrate that our method outperforms several state-of-the-art HSI SR methods. The code will be available at https://github.com/YoeTon/CTUNet.**

***Index Terms*— Convolution neural network (CNN), hyperspectral image (HSI), super-resolution (SR), transformer, unfolding mechanism.**

## I. Introduction

HYPERSPECTRAL remote sensing is the process of collecting reflectance information from ground objects in hundreds of narrow continuous spectral channels to form 3-D hyperspectral images (HSIs) [1], [2]. On the one hand, a 3-D HSI consists of a set of 2-D grayscale images containing abundant texture details, which corresponds to the morphology, distribution, and reflection information of the ground objects. On the other hand, the spectral response curve corresponding to each pixel precisely characterizes the composition of different substances [3]. Consequently, HSIs are widely used for detection and target analysis in both 2-D spatial and 1-D spectral space, with significant applications in mineral mapping [4], feature classification [5], target detection [6], object tracking [7], [8], and ecological assessment [9]. However, due to the limited energy received in each band, the precise spectral information of HSIs comes at the corresponding cost of lower spatial resolution, which makes them inadequate for practical applications [10], [11].

HSI super-resolution (HSI SR) is an effective and low-cost image post-processing technology, aiming to generate a high-resolution (HR) HSI from its low-resolution (LR) counterpart. In general, HSI SR can be categorized into single HSI SR approaches and multiframe HSI SR methods [12]. Nevertheless, multiframe-based methods usually require multiple LR images of the same scene [13], which brings a significant challenge in spatial registration in real scenes. In this regard, single HSI SR has gained increased attention due to its flexibility and applicability without additional data.

Given the limited information provided by a single LR HSI, it is necessary to address the ill-posed SR problem by exploring image priors from large external image datasets. Conventional knowledge-driven methods mainly focus on the utilization of effective spatial and spectral statistical distributions such as low rank, self-similarity, and total variation to describe edges and textures [14], [15]. However, these methods usually require complex manual priors that need to match the nature of the images.

In the past decade, with the emergence of large-scale image datasets and rapid growth in computational power, deep learning-based methods, which are a data-driven approach, have demonstrated superior performance against traditional methods [16]. A straightforward way to accomplish the single HSI SR is to apply natural or gray image SR methods [17]. However, HSI usually contains hundreds of bands reflecting a wide range of ground information, and there is a strong correlation between these bands, which leads to increased difficulty in feature extraction and is prone to redundancy in information utilization. To address the above problem, existing approaches focus on exploring richer spectral information with remarkable progress in novel network modules or learning

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Yao Tang, Yajie Li, and Yi Xiao are with the School of Geodesy and Geomatics, Wuhan University, Wuhan 430079, China (e-mail: tyvitor@whu.edu.cn; l_anyways@163.com; xiao_yi@whu.edu.cn).

Jie Li is with the School of Geodesy and Geomatics and the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430072, China (e-mail: aaronleecool@whu.edu.cn).

Linwei Yue is with the School of Geography and Information Engineering, China University of Geosciences, Wuhan 430074, China (e-mail: yuelw@cug.edu.cn).

Xinxin Liu is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China (e-mail: liuxinxin@hnu.edu.cn).

Qiangqiang Yuan is with the School of Geodesy and Geomatics, the Collaborative Innovation Center of Geospatial Technology, and the Key Laboratory of Polar Environment Monitoring and Public Governance, Ministry of Education, Wuhan University, Wuhan 430072, China (e-mail: yqiang86@gmail.com).

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strategies, including 3-D convolution [18], channel attention mechanism [19], spectral angle mapper (SAM) loss [20], and group convolution [21]. However, these designs struggle to exploit spectral correlations and capture long-range dependencies. Therefore, transformer technique and its variants [22] are getting more attention recently to implement high-performance HSI SR due to the powerful global modeling capabilities.

Although deep learning-based methods can lead to excellent reconstructions, the "black-box" has confused many scientists about the interpretability of such methods. Moreover, a simple mapping from LR images to HR images leads to the significant domain gap in the modeling process while neglecting the characterization of the degradation process. That is to say, regardless of the output, the working principle of the SR process is difficult to understand. In order to achieve higher performance, the structure of the network becomes more complex and confusing. Moreover, the number of parameters will inevitably increase, which poses significant obstacles in practical applications. Therefore, to ensure the interpretability and reduce the number of parameters in solving, some works [23], [24] unfold the degradation process of HR HSIs and replace certain modules with neural networks. By iteratively seeking the optimal solution, they ultimately achieve satisfactory results. However, these methods often overlook the relationships between features generated at each iteration, and the performance of the embedded neural networks are not thoroughly explored, which limit the reconstruction performance.

Similar to the aforementioned methods, we also combine the knowledge-driven and deep-learning-based methods to grasp the mutual prompting properties of both the methodologies. We propose a novel CNN-transformer embedded unfolding network (CTUNet) for HSI SR task, in which an iterative algorithm with alternating optimization to search for the reconstruction results and an hyperspectral image spatial–spectral prior network (HSSPN) for exploring deep priors is embedded into the iterative process. This end-to-end training model not only incorporates degradation information, allowing the underlying working principle to be explained to a certain extent, but also the embedding of prior networks makes the network more efficient and high performance. Not treating features generated at each iteration independently like the aforementioned methods, inter-stage information fusion (ISIF) module is utilized in our method to learn cross-stage latent spatial–spectral priors. Besides, HSSPN in the framework unites convolution neural network (CNN) and transformer and achieves the integration of local and global information, focusing on the multiscale information in the spatial dimension and similarity in the spectral dimension of HSIs. Specifically, multiscale self-calibrated convolution (MSSC) improves on the classic self-calibrated convolution and also learns features on a larger scale. Egde-guided transformer block (EGTB) achieves long-distance information mining, and the large receptive field in the feedforward network also enhances network performance. It is worth mentioning that we incorporate edge feature representation into the transformer module, providing edge priors for high-quality reconstruction, thus restoring more details. In summary, the main contributions of this work are threefold.

1) The end-to-end trainable unfolding network CTUNet based on maximum a posterior (MAP) framework is proposed. This interpretable multistage network could embed SR networks, bridging the gap between knowledge-driven and deep learning-based approaches. In addition, ISIF module aggregates features from neighboring iteration stages, which effectively improves the image characterization capability.
2) HSSPN is designed for the benchmark bicubic degradation, where MSSC and EGTB effectively combined to extract local and global features, achieving efficient spatial–spectral representation.
3) Extensive analysis experiments and ablation experiments demonstrate the effectiveness of the proposed method. Experiments on three benchmark datasets with different degradation models testify that the proposed approach can achieve state-of-the-art performance.

The remainder of this article is organized as follows. In Section II, we introduce the existing HSI SR methodology and the work related to our approach. In Section III, we describe the proposed CTUNet in detail. Section IV presents the implementation details and results of the experiments, as well as the analysis of the model structure. Finally, conclusion is given in Section V.

## II. Related Works

### A. Knowledge-Driven HSI SR

Knowledge-driven methods for HSI SR usually consider the LR images as prior knowledge and regularize the solution space using prior assumptions. In an early work, Akgun et al. [25] propose a model that represents the hyperspectral observations from different wavelengths as weighted linear combinations of a small number of basis image planes and then the spectrum of the observed scene is reconstructed by fusing information from multiple observations and spectral bands. In [26], principal component analysis (PCA) is utilized in motion estimation and image reconstruction, in which different image resolution enhancement techniques are applied to different groups of components. Huang et al. [14] propose an SR approach with unknown blurring by imposing the low-rank model with predefined spectral subspace and group sparse model to utilize the shared spatial structure across all spectral bands. Sparse representation, total variation, and non-local self-similarity-based methods can also effectively utilize image features and details. Li et al. [27] propose a novel framework that combines spectral mixture analysis and spatial–spectral group sparsity, combining sparsity and non-local self-similarity. This model not only maintains the spectral consistency but also preserve abundant image details. He et al. [15] utilize 3-D total variation to characterize the local spatial–spectral smoothness across all hyperspectral bands. Wang et al. [28] propose a tensor-based approach to exploit the three intrinsic characteristics of HS image, i.e., the global correlation across spectral domain, the non-local self-similarity across spatial domain, and the local smooth structure across both spatial and spectral domains, and achieve superior reconstruction performance. However, this knowledge-driven

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approaches increasingly require complex priors and time costs to meet good performance.

### B. Deep Learning-Based HSI SR

Due to the significant success of deep learning technology in the field of SR, neural networks, represented by CNN, have been widely applied in HSI SR. Different from directly applying networks such as SRCNN or EDSR for natural image SR to HSIs, Mei et al. [18] consider the rich spectral information of HSIs and innovatively propose 3-D full convolutional neural network, which fully utilizes the spatial context of neighboring pixels and the spectral correlation of neighboring bands. Li et al. [29] introduced an SDCNN for spectral difference mapping learning to preserve spectral information, combined with a spatial constraint to improve spatial details. However, these early methods often corresponded to higher computational complexity and memory loss. For these cases, strategies such as separable 3-D convolution [30], spectral grouping, parameter sharing [21], and recurrent networks [20] have been introduced in HSI SR.

Li et al. [20] propose a grouped deep recursive residual network with grouped recursive module embedded into the global residual structure, and the SAM loss is combined to optimize the network. A classic spectral grouping network, proposed in [21], utilizes a group convolution and progressive upsampling framework, greatly reducing the size of the model, and design a spatial–spectral prior block which can fully exploit the spatial information and the correlation between the spectra of the hyperspectral data. Then, Liu et al. [31] introduce a second-order spectral attention mechanism constructed from feature covariance statistics for better mining spectral priors. In [32], a novel recursive network combining 2-D convolution and separated 3-D convolution solves the problem of structural redundancy and enhances the learning ability of spatial–spectral domain.

Although CNN-based methods achieve satisfactory reconstruction results, their inadequacy in exploring long-range dependencies has led scientists to pay attention to transformer, which enable global information extraction. Transformer is originally proposed by Vaswani et al. [33] for natural language processing (NLP). Due to its excellent ability to extract global features, it has demonstrated its superiority and feasibility in many image restoration works [34], [35], [36], [37]. Therefore, more literature has designed transformer-based models or models combining transformer and CNN for implementing HSI SR task. Liu et al. [38] propose an interactive network called Interactformer to interact with global and local features extracted by transformer and 3-D CNN branches, but it also caused serious memory loss. In order to achieve smaller computational complexity and higher performance, Long et al. [22] propose a dual self-attention swin transformer network (DSSTSR) that utilizes the ability of the swin transformer in the spatial representation of both global and local features and learns spectral sequence information from adjacent bands of HSI. Although these methods can achieve excellent performance, most of them are empirically designed, which are not explainable. In addition, more complex network structures and more model parameters have become a trend, leading to an increase in the burden on computers.

In order to effectively utilize the degradation principle during image formation, some works [39] focus on modeling spatial degradation induced by sensors on the images, in addition to using neural networks in the main part. By utilizing the spatial response knowledge of the sensor, an SR method tailored to existing satellites has been developed. However, it is evident that although this method performs well on one specific sensor, its generalization ability is insufficient in more scenarios, and it also requires the use of specific parameters that cause spatial degradation during the optical imaging process. Therefore, in order to complete the SR task for images from different sensors, we need to find an approach that can maximize the use of image degradation characteristics.

### C. Unfolding Methods for HSI SR

Due to the effectiveness of the methods that couple knowledge-driven and data-driven ideas in addressing image inverse problems, including image SR, they have received widespread attention [40], [41]. The unfolding strategy is a mainstream interpretable SR framework that can leverage both optimization model and neural network. A typical deep unfolding process usually includes constructing an iterative solving algorithm and embedding learnable neural networks. Specifically, these methods utilize the variable splitting techniques, such as the half-quadratic splitting (HQS) algorithm [42] and alternating direction method of multipliers (ADMM) algorithm [43], to transform SR problem into two independent subproblems, consisting of a data fidelity term that guarantees the solution conforms to the degradation process and a prior regularizer that enforces desired properties of the output [44]. The unfolding methods not only have a certain degree of interpretability but can also produce quite superior results with fewer parameters and iterations. On this basis, different from the deep plug-and-play methods or deep denoising-based methods [45], Zhang et al. [46] propose a deep unfolding method for image restoration, jointly optimizing the two subproblems in an end-to-end manner. Afterward, Mou et al. [47] integrate a gradient estimation strategy into the gradient descent step of the proximal gradient descent (PGD) and design a deep generalized unfolding network. In [48], a deep unfolding network is presented by incorporating a new regularization module and putting the regularization part, which is deducted by using the regularization by denoisng (RED), before the data fitting module. Recent works have introduced the deep unfolding strategy into HSI SR. To the best of our knowledge, Liu et al. [23] innovatively propose a deep unfolding network (EUNet) for single HSI SR. Specifically, the optimization process of the SR prior-driven MAP framework is unfolded into an interpretable multistage network, which inherits the advantages of deep learning-based image SR and explicitly imposes the degradation model constraint. Unlike the regularization term in EUNet which only uses simple 2-D convolution and attention mechanism, Wang et al. [49] cleverly embed transformer, 3-D convolution, and 2-D convolution into the optimization process, achieving great results, although this also brings about significant memory loss.

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## III. Methodology

The proposed CTUNet learns an end-to-end mapping between the observed LR HSI and the corresponding HR HSI in an iterative manner. Unlike directly establishing a prior learning framework through neural networks, considering that the rich domain knowledge of HSIs can better understand the ill-posedness of SR problems from a mechanistic perspective, we start from the degradation model of HR HSI and introduce a CTUNet that can effectively achieve the representation of spatial and spectral features.

### A. Degradation Model

Throughout this article, denote an observed LR HSI as $\mathcal{Y} \in \mathbb{R}^{h\times w\times B}$ contains $B$ bands with spatial size of $h \times w$. Correspondingly, the HR HSI to be constructed is denoted as a third-order tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$, where $H = s \times h$, $W = s \times w$ and $s$ represents the scale factor. According to [50], the blur degradation matrix of each spectral band can be considered independent, thus the degradation model of each band can be given by

$$\mathcal{Y}_b = (\mathcal{X}_b \downarrow_s \otimes K_b) + N_b \tag{1}$$

where $\mathcal{Y}_b$ represents the $b$th band of the observed LR HSI, and its corresponding HR HSI band is $\mathcal{X}_b$. $\downarrow_s$ and $K_b$ represent a downsampling operation and a blur kernel which achieve the degradation of HSI. $N_b$ is the additive independent noise.

The degradation matrices corresponding to all bands can be represented by the matrices acting on the entire image, so by unfolding 3-D data cubes into 2-D matrices, the HSI degradation model can be written as

$$Y = \mathrm{HDX} + N \tag{2}$$

where $Y \in \mathbb{R}^{hw\times B}$ and $X \in \mathbb{R}^{HW\times B}$ are the 2-D representations of LR HSI and HR HSI, respectively. $H \in \mathbb{R}^{hw\times hw}$ denotes the blurring matrix and $D \in \mathbb{R}^{hw\times HW}$ represents the bicubic downsampling operation acting on HR HSI. And $N$ is the stack of additive noise in all bands. The degradation model conveys that the LR HSI is a bicubicly downsampled, blurred, and noisy version of the HR HSI.

### B. Iteration Mechanism and Unfolding Optimization

Obviously, reconstructing an HR HSI $X$ from the observed LR HSI is an ill-posed inverse problem, which means that the MAP method is a crucial approach for generating the estimated HR HSI $\hat{X} \in \mathbb{R}^{H\times W\times B}$. Thus, according to MAP framework, we can maximize the posterior probability of $\hat{X}$ as follows:

$$\hat{X} = \underset{X}{\operatorname{argmax}}\, P(X|Y) \propto P(Y|X) \cdot P(X). \tag{3}$$

To simplify the inference process, (3) can be written in an equivalent form by applying a negative logarithmic transformation

$$\hat{X} = \underset{X}{\operatorname{argmin}} -\log P(Y|X) - \log P(X) \tag{4}$$

where $P(Y|X)$ denotes the conditional probability of the degraded version $Y$, and $\log P(X)$ is the prior probability of $X$. Thus, $\log P(Y|X)$ and $\log P(X)$ can represent the data fidelity and regularization terms, respectively. Since the data fidelity is usually defined as $\mathcal{L}_2$ norm, (4) can be further reformulated as the following energy function:

$$\hat{X} = \underset{X}{\operatorname{argmin}} \frac{1}{2\sigma^2}\|Y - HDX\|_2^2 + \lambda\Phi(X) \tag{5}$$

where $\lambda$ is a hyperparameter to weight the regularization term $\Phi(X)$ and $\sigma$ is used to control the noise level. The first item is the data fidelity, which ensures data consistency and conforms to the degradation process. The second term stands for the regularization term, describing the prior knowledge of $X$. The optimization problem (5) can be effectively solved by some variable splitting techniques, such as the HQS algorithm and ADMM algorithm. By employing the mentioned unfolding strategy and introducing an auxiliary variable $Z$ constrained to be equal to $DX$, (5) can be written as follows:

$$(X, Z) = \underset{X,Z}{\operatorname{argmin}} \frac{1}{2\sigma^2}\|Y - HZ\|_2^2 + \lambda\Phi(X) + \frac{\mu}{2}\|Z - DX\|_2^2 \tag{6}$$

where $\mu$ is the penalty parameter that constraints the similarity between $Z$ and $X$. Then, (6) can be alternatively solved by two independent subproblems

$$\begin{cases} Z_t = \underset{Z}{\operatorname{argmin}} \frac{1}{2\sigma^2}\|Y - HZ_{t-1}\|_2^2 + \frac{\mu_t}{2}\|Z_{t-1} - DX_{t-1}\|_2^2 \\ X_t = \underset{X}{\operatorname{argmin}} \frac{\mu_t}{2}\|Z_t - DX_{t-1}\|_2^2 + \lambda\Phi(X_{t-1}) \end{cases} \tag{7}$$

where $t$ means the $t$th iteration, while $t \in [1, T]$. The $Z$-subproblem and the $X$-subproblem are, respectively, associated with the data fidelity term used to constrain data consistency and the regularization term used to acquire prior information. Based on such an iterative process, the two subproblems can be solved using two different neural network modules. Fig. 1 illustrates the overall architecture of CTUNet with two core modules, namely the fidelity layer (FL) and the prior-learning layer (PL). The FL assists in updating the auxiliary variable $Z$, and then the PL takes $Z$ as input to generate the reconstruction result $X$ at each stage. After $T$ iterations, an optimal solution will be produced as the final reconstructed SR image. Sections III-C and III-D will describe how to solve these two subproblems.

### C. Fidelity Layer

In the deep unfolding framework, the FL aims to solve the $Z$-subproblem, which effectively reduce the blur and distortion [23], thus outputting an optimal solution $Z_t$. To avoid computations involving large matrices and enable back propagation of gradients, gradient descent based method is chosen to compute an approximate solution. For the sake of simplicity and esthetics of the formula, the $Z$-subproblem can be rewritten in an equivalent form

$$Z_t = \underset{Z}{\operatorname{argmin}} \frac{1}{2}\|Y - HZ_{t-1}\|_2^2 + \frac{\mu_t\sigma^2}{2}\|Z_{t-1} - DX_{t-1}\|_2^2. \tag{8}$$

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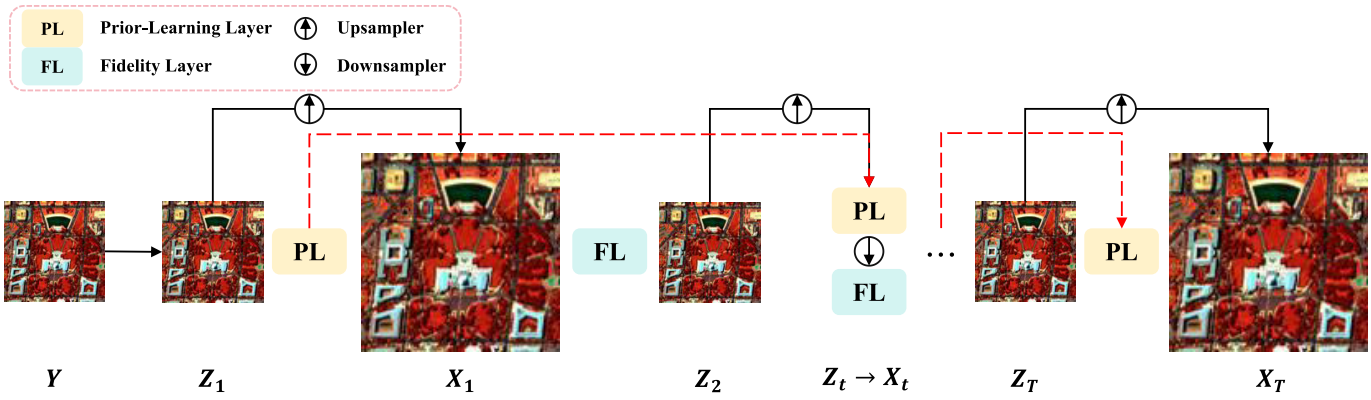



Fig. 1. Overall architecture of our proposed CTUNet for HSI SR.

Following [44], taking $Z_{t-1}$ as the initial point, we do a single step of gradient descent to update $Z$:

$$Z_t = Z_{t-1} - \eta \nabla f(Z_{t-1}) \tag{9}$$

where $\eta$ is the step size, and $f(\cdot)$ is the objective function of the optimization problem with the following form:

$$f(Z) = \frac{1}{2}\|Y - HZ\|_2^2 + \frac{\mu_t \sigma^2}{2}\|Z - DX_{t-1}\|_2^2. \tag{10}$$

By computing the gradient of $f(Z)$ and then simply substituting it into (9), the final solution of $Z$ can be obtained as follows:

$$\begin{aligned} Z_t &= Z_{t-1} - \eta_t\left[H^T(HZ_{t-1} - Y) + \alpha_t(Z_{t-1} - DX_{t-1})\right] \\ &= \bar{H}Z_{t-1} + \eta_t H^T Y + \eta_t \alpha_t D X_{t-1} \\ &= \mathrm{FL}(Z_{t-1}, X_{t-1}, Y, H, D, \alpha_t, \eta_t) \end{aligned} \tag{11}$$

where $\alpha_t = \mu_t \sigma^2$, $\bar{H} = [(1 - \eta\alpha)I - \eta H^T H]$. FL represents the FL, which plays an important role in promoting data consistency. This layer is responsible for updating the auxiliary variable $Z$, which will serve as input data during the subsequent solving process of the $X$-subproblem. Note the input data and initialization in (11), $H$ and $D$ are the input to FL as the blur matrix and downsampling matrix, respectively, which imposes an explicit degradation constraint on the solution. $\alpha$ and $\eta$ will be adaptively generated using two implicit fully connected networks. And $Z_1$ is initialized as $H^T Y$. While in our experiments, $H$ and $H^T$ are set as identity matrices, while $D$ and $D^T$ are implemented using bicubic interpolation.

### D. Prior Learning Layer

During the iteration process, after obtaining the solution of $Z$ in the FL, the PL is designed to solve the $X$-subproblem, which can be regarded as searching for high-quality forms of the reconstructed HSIs. Due to the inadequacy of handcrafted priors to fully explore the prior information of HSIs, instead of using simple model-based methods, we opt to learn sufficient prior knowledge from the training data through a powerful deep neural network and utilize these features to solve the SR problem. Thus the $X$-subproblem can be solved by

$$\begin{aligned} X_t &= \underset{X}{\operatorname{argmin}} \frac{\mu_t}{2}\|Z_t - DX_{t-1}\|_2^2 + \lambda\Phi(X_{t-1}) \\ &= \underset{X}{\operatorname{argmin}} \frac{1}{2\left(\sqrt{\lambda/\mu_t}\right)^2}\|Z_t - DX_{t-1}\|_2^2 + \Phi(X_{t-1}) \\ &= \mathrm{PL}(Z_t, \beta_t) \end{aligned} \tag{12}$$

where PL denotes the PL implemented by a carefully designed network, while its input consists of the parameters $\beta_t = \sqrt{\lambda/\mu_t}$ generated by two fully connected layers and the output of FL $Z$.

As it is well known, the complementary abilities of CNN in extracting local features and transformer in extracting global features enable them to jointly explore the 3-D spatial–spectral features of HSIs comprehensively. Building upon this, HSSPN is proposed to solve the $X$-subproblem, playing a crucial role in probing the intrinsic structure of $X$. Specifically, HSSPN contains three key modules, namely MSSC, EGTB, and ISIF. These modules fully leverage the multiscale features and the high spectral correlation of HSIs, enabling the learning of intricate spatial–spectral prior.

### E. Hyperspectral Spatial–Spectral Prior Network

In this section, the proposed HSSPN is presented in detail. As shown in Fig. 2, HSSPN mainly includes four parts: shallow feature extraction, deep feature extraction consisting of several local–global feature representation block (LGFRB), ISIF, and reconstruction. Specifically, a 2-D convolution layer is first applied to the LR input $Z_t$ generated in FL, which maps the input to the feature space with more informative image gradients. Then, the features are fed into deep feature extraction module where a cascade of LGFRBs are utilized to capture higher-level global and local information. The LGFRB contains an MSSC module and an EGTB connected in sequence. In detail, the MSSC first explores the internal representations of input features in a multiscale manner, which are then fed into EGTB to explore global features leveraging spectral correlation and edge information of HSIs. Then, shallow and deep features are aggregated with a global skip connection to reconstruct the HR image. After that, to enhance the representation of the current stage by leveraging deep

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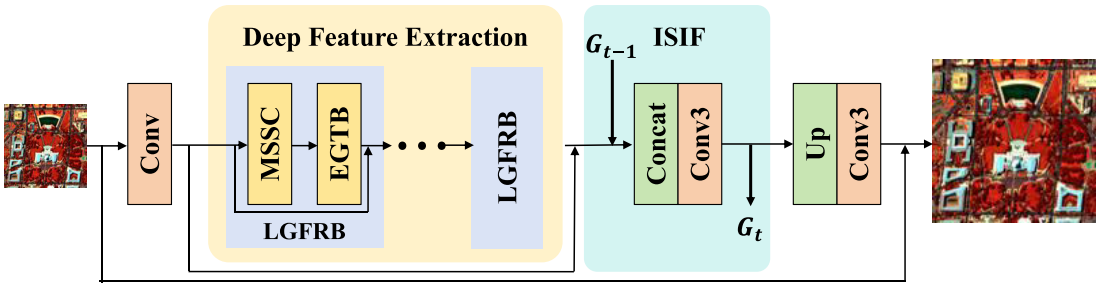



Fig. 2. Architecture of HSSPN for PL.

features from the previous stage, an ISIF module follows, establishing interaction between adjacent stages. In the reconstruction module, a PixelShuffle operation and a convolution layer are utilized for upsampling and reducing the number of channels to the same as the bands of the input HSIs.

*1) Multiscale Self-Calibrated Convolution:* Ordinary self-calibrated convolutions (SCConvs) with attention mechanism can help CNNs generate more discriminative representations by explicitly incorporating richer information [51]. Inspired by this, we propose MSSC that can effectively expand the receptive field at each spatial position, utilizing spatial relationships among adjacent pixels. This multiscale attention design enables the network to focus on the continuity of LR and HR features of the same object. As shown in Fig. 3, first, the input features are expanded threefold in channel number through a $3 \times 3$ convolution layer and they are uniformly divided into three parts along the channel dimension and inflow into three branches. The middle branch adopts the simplest structure consisting of a $3 \times 3$ convolution layer followed by a ReLU activation function and another convolution, ensuring preservation of the original features of the image. The bottom branch employs an attention mechanism, where the features processed through a $3 \times 3$ depthwise convolution are multiplied by attention weights, which are generated by a $1 \times 1$ convolution followed by a sigmoid function. The structures of the top and bottom branches are similar, with two minor modifications. The feature extraction in this branch is conducted at a larger scale, hence an upsampling and downsampling operation implemented by a transposed convolution and a convolution with stride 2, respectively, are applied before and after this branch, respectively. Additionally, a dilated depthwise convolution is employed to enlarge the receptive field of the network. Finally, the output features of three branches are concatenated and then fed into a $1 \times 1$ convolution layer to generate the final hierarchical spatial and spectral features. And the final output of MSSC can be obtained by adding these features to the input features. It is evident that our MSSC can adaptively encode both spatial and channel-wise interaction information for each pixel in the region, thereby generating more hierarchical information. After a short skip connection, the deep features of MSSC module can be formulated as follows:

$$F_{\text{MSSC}} = F_{\text{in}} + \text{Conv}_{1\times1}\left(\text{Cat}\left[F_{\text{up-down}}, F_{\text{main}}, F_{\text{SC}}\right]\right) \quad (13)$$

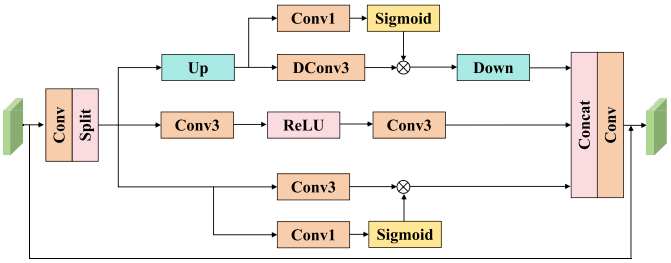



Fig. 3. Architecture of MSSC.

where $F_{\text{in}}$ denotes the input feature of MSSC, and $F_{\text{up-down}}$, $F_{\text{main}}$, and $F_{\text{SC}}$ represent the output of the upper, middle, and lower branches of MSSC, respectively.

*2) Edge-Guided Transformer Block:* Although the module MSSC can fully explore local features for spatial reconstruction, its insufficient mining of non-local features limits the SR performance. And our proposed EGTB effectively achieves deep exploration of global contextual information, which is an improvement of the vision transformer [52]. The details of EGTB are illustrated in Fig. 4, including two key parts: edge-guided channel self-attention (EGSA) and wide attention block (WAB). An EGTB can be mathematically denoted as EGTB$(\cdot)$ and computed as

$$\begin{aligned} F_{\text{SA}} &= \text{EGSA}(F_{\text{MSSC}}) + F_{\text{MSSC}} \\ F_{\text{EGTB}} &= \text{WAB}(F_{\text{SA}}) + F_{\text{SA}}. \end{aligned} \quad (14)$$

According to [53], the attention maps across different positions are almost uniform, and it suggests significant redundancy in computing attention maps for each point in the non-local module. Additionally, considering the high similarity between spectral bands in HSIs, we propose a simplified channel self-attention for achieving lightweight global information extraction shown in Fig. 4(a). Furthermore, for HSI SR task, the introduction of high-frequency information is helpful to restore the texture details [54]. Therefore, EGSA adopts an optical flow inspired edge body generation mechanism (EBGM) [55] to generate the edge prior, as shown in Fig. 4(c). We first use bilinear interpolation to downsample the original feature map $F$, and then perform its inverse process to restore it to the same size as the original feature map and concatenate the two features. These features are then input into a convolution layer to predict the flow field, and the body part $F_{\text{body}}$ is obtained after deformation. Finally, the edge prior $F_{\text{edge}}$ is obtained through subtraction of the original feature map $F$ and its body part $F_{\text{body}}$. Returning to EGSA, the $Q$ and

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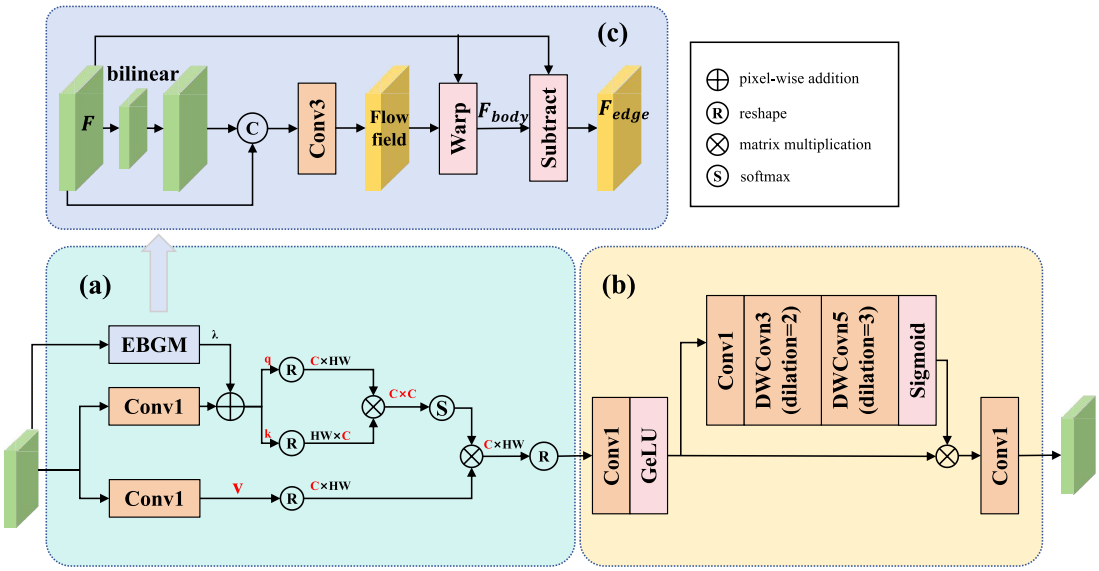



Fig. 4. Architecture of EGTB, which consists of two main modules. (a) EGSA and (b) WAB. In EGSA, (c) EBGM is utilized to generate the edge prior.

$K$ are obtained by adding the output of EBGM with the same $1 \times 1$ convolution, where the edge information is controlled by a fixed weight $\alpha$. Then, the similarity map $S \in \mathbb{R}^{C\times C}$ can be generated from the matrix dot product between $Q$ and $K$. $V$, obtained from a $3 \times 3$ convolution, is multiplied with $S$ to output the final features. The calculation process can be described as follows:

$$\begin{aligned} Q, K &= \text{Conv}_{1\times1}(F_{\text{MSSC}}) + \alpha\text{EBGM}(F_{\text{MSSC}}) \\ V &= \text{Conv}_{1\times1}(F_{\text{MSSC}}) \\ F_{\text{SA}} &= F_{\text{MSSC}} + \text{softmax}\left(\frac{QK_t}{\sqrt{d_k}}\right)V. \end{aligned} \tag{15}$$

It is worth noting that the multihead strategy has been employed, resulting in a significant reduction in computational workload.

As for WAB shown in Fig. 4(b), many works have proved that introducing large receptive fields in the attention branches improves the SR effect [56], [57]. Due to the limited capacity of simple MLP-based feed-forward networks to fully explore deep features and the increase in computational complexity, we designed WAB with a vast receptive field. We first perform a $1 \times 1$ convolution and a Gaussian error linear unit (GELU) activation to the input feature. In the attention module, we sequentially utilize $1 \times 1$ convolution, $3 \times 3$ depthwise convolution with dilation of 2, and $5 \times 5$ depthwise convolution with dilation of 3 to generate attention weights, which enables the receptive field to reach 17. After an element-wise product, another $1 \times 1$ convolution layer projects the deep feature. At last, local residual learning further improves the information flow and network representation ability.

*3) Inter-Stage Fusion Module:* Considering that most deep unfolding network methods are negatively affected by information loss due to the feature-to-image transformation at the end of each stage [47], we design ISIF module to broadcast useful information from the previous stage. It first concatenate the deep features in the current stage with the fused features $G_{t-1}$ from the previous stage. This data is then fed into a $1 \times 1$ convolution layer for fusion. Subsequently, it inflows into the reconstruction module, while the fused features $G_t$ are also fed into the next stage. The ISIF helps the information propagation across stages and avoid redundant computation.

### F. Loss Function

For the restoration task of HSI with rich 3-D information, spatial reconstruction and spectral consistency are the two most critical issues. The $\mathcal{L}_1$ loss is the most commonly used loss function in SR tasks, while the SAM loss can effectively measure the spectral similarity between the reconstructed images and the ground-truth images. Therefore, the $\mathcal{L}_1$ loss and SAM loss are used for jointly optimize the network. In addition, since gradient information can serve as a powerful tool to enhance the sharpness of SR images, we also adopt a gradient constraint for minimizing the gradient loss. Hence, the final objective loss of the proposed network can be represented as a weighted sum of these three losses

$$\begin{aligned} L_1 &= \frac{1}{N}\sum_{i=1}^{N}\|X_i - \hat{X}_i\|_1 \\ L_{\text{SAM}} &= \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{X_i \cdot \hat{X}_i}{\|X_i\|_2\|\hat{X}_i\|_2}\right) \\ L_{\text{gra}} &= \frac{1}{N}\sum_{i=1}^{N}\|M(X_i) - M\left(\hat{X}_i\right)\|_1 \\ L_{\text{total}} &= L_1 + \lambda_s L_{\text{SAM}} + \lambda_g L_{\text{gra}} \end{aligned} \tag{16}$$

where $N$ is the number of LR–HR patch pairs. $M$ computes the gradient value along horizontal, vertical, and spectral dimensions of image. $\lambda_s$ and $\lambda_g$ control the weight of SAM loss and gradient loss, respectively. In our experiment, we empirically set $\lambda_s = 0.1$, $\lambda_g = 0.01$.

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## IV. Experiments

### A. Experimental Settings

*1) Datasets:* We evaluate the proposed method on four public datasets from different sensors, including Pavia Center and University,[1] Chikusei,[2] and Houston.[3] The Pavia Center and Pavia University datasets were acquired by the ROSIS captured over an urban area in Italy. Pavia Center covers wavelengths ranging from 430 to 860 nm, with a total of 115 bands, some of which removed due to water vapor absorption and noise. Finally, Pavia Center forms a valid image of size $1096 \times 715 \times 102$, which includes left and right parts. In our experiments, four sub-images of size $128 \times 128 \times 102$ at the bottom are used for test and the remaining parts are segmented into patches of size $64 \times 64 \times 102$ pixels for training and validation, with the left and right sections being processed separately. Pavia University has 103 bands with a spatial size of $610 \times 340$. We select a region with spatial pixels of $200 \times 200$ as the original image for real experiments.

The Chikusei dataset was acquired using the Headwall Hyperspec-VNIR-C imaging sensor in agricultural and urban areas of Chikusei City, Japan, with a ground sampling distance of 2.5 m, and the scene consists of $2517 \times 2335$ pixels, covering 128 bands from 363 to 1018 nm. Four non-overlapping sub-images of size $512 \times 512 \times 128$ pixels from the top area were selected to construct the test set.

The Houston dataset was captured by the ITRES CASI 1500 spectral imager, covering the University of Houston campus and its surrounding urban area, with a ground sampling distance of 1 m. The imagery consists of 144 spectral bands in the 380–1050 nm region. Eight non-overlap images of size $128 \times 128 \times 48$ at the bottom are chosen as the test data.

In our experiments, aside from the aforementioned test portion, the remaining regions are used to construct the training set, of which 10% is used for validation. Specifically, the remaining part was divided into patches of size $64 \times 64$ in spatial dimensions, with some overlap between adjacent patches to ensure an adequate number of training images. These images are then downsampled by bicubic interpolation at $\times 2$ or $\times 4$ level to simulate LR images, which are used to construct LR–HR image pairs.

*2) Evaluation Metrics:* Three mainstream evaluation criteria are utilized to assess HSI SR tasks, both in spatial and spectral domains. To validate the quality of spatial reconstruction, we select peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). The calculation of PSNR considers the mean-squared error (mse) between the HR images and the SR images and converts it into decibel (dB) units, providing an intuitive way to measure the clarity and accuracy of the reconstructed images. SSIM can evaluate the similarity and structural preservation between images, capturing perceptual quality differences in images and effectively assessing image details. In the spectral dimension, SAM is used to assess the reconstruction quality of each spectrum at the pixel level, measuring the similarity between two spectral vectors by calculating the angle between them. Larger PSNR and SSIM values mean better spatial reconstructed results, while smaller values of SAM indicate less spectral distortion.

*3) Competing Methods:* We compare the performance of the proposed CTUNet with six representative approaches, including Bicubic, GDRRN [20], SSPSR [21], GELIN [58], MSDformer [59], and EUNet [23]. Bicubic is a classic and simple interpolation method and GDRRN is a 2-D convolution-based method that employs a recursive strategy. SSPSR and GELIN are both state-of-the-art methods that group the input image. MSDformer is an advanced network that combines CNN and transformer. And EUNet is also an unfolding method. We utilize the open-source codes of these networks, following the corresponding parameter settings, and train them on the same dataset until convergence.

*4) Implementation Details:* We implements and trains the proposed CTUNet using the Pytorch library on two Nvidia RTX 2080Ti GPUs, with a scale factor $\times 2$ and $\times 4$. During training, the ADAM optimizer [60] is utilized with an initial learning rate of $1e{-}4$, and the learning rate is decayed by a factor of 1/2 every 150 epochs. In our experiments, the iteration number $T$ is set to 4 and the number of LGFRB $K$ is set to 3. Considering the scale of the network, 32 LR patches are randomly cropped from training images as input for each training batch and are augmented by randomly flipping vertically and rotating 90°, 180°, and 270° to alleviate the overfitting and improve the generalization ability of the network.

### B. Comparisons to State of the Arts

*1) Quantitative Results:* Table I shows the comparative results of the proposed method and six state-of-the-art methods designed for HSI SR on three datasets, with the best results highlighted in bold and the second best results highlighted in underline. As seen, our CTUNet outperforms the other comparative methods, demonstrating the effectiveness of the design of the proposed approach in this article.

Specifically, the interpolation method Bicubic naturally achieves the poorest results despite not requiring any training sample. GDRRN is limited by model capacity, although it achieved relatively good results in considering spectral correlations, its performance struggled to achieve further improvement. SSPSR and GELIN, as networks meticulously designed for HSI SR tasks, adopt spectral grouping strategies, achieving excellent reconstruction results from both spatial and spectral perspectives. MSDformer effectively combines the advantages of CNN and transformer, leveraging the redundancy of information between spectral groups, obtaining nice results in spatial reconstruction and spectral recovery. EUNet is an unfolding network that leverages the principles of HSI degradation in physics, embedding a simple network structure in establishing the framework for HSI restoration, thus achieving outstanding reconstruction accuracy under lightweight network settings. However, overall, our method outperforms others on three different datasets, almost

[1]https://ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[2]https://www.sal.t.u-tokyo.ac.jp/hyperdata/

[3]https://hyperspectral.ee.uh.edu/?page_id=1075

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TABLE I
QUANTITATIVE COMPARISONS OF DIFFERENT METHODS ON PAVIA CENTER, CHIKUSEI, AND HOUSTON FOR SCALE FACTORS ×2 AND ×4

| Dataset | Scale | Metric | Bicubic | GDRRN [20] | SSPSR [21] | GELIN [58] | MSDformer [59] | EUNet [23] | CTUNet |
|---|---|---|---|---|---|---|---|---|---|
| Pavia Center | 2 | PSNR | 30.879 | 33.478 | 34.853 | 34.820 | 35.138 | 34.898 | **35.970** |
| | | SSIM | 0.9542 | 0.9742 | 0.9806 | 0.9807 | 0.9817 | 0.9813 | **0.9842** |
| | | SAM | 3.878 | 3.338 | 3.158 | 3.048 | 3.019 | 3.005 | **2.906** |
| | 4 | PSNR | 26.190 | 27.213 | 27.577 | 27.543 | 27.903 | 27.437 | **28.044** |
| | | SSIM | 0.8561 | 0.8921 | 0.9064 | 0.9038 | 0.9110 | 0.9005 | **0.9152** |
| | | SAM | 5.537 | 4.955 | 4.842 | 4.993 | 4.878 | 4.896 | **4.728** |
| Chikusei | 2 | PSNR | 36.758 | 39.227 | 40.735 | 41.067 | 41.288 | 40.740 | **41.371** |
| | | SSIM | 0.9849 | 0.9915 | 0.9940 | 0.9944 | 0.9947 | 0.9941 | **0.9954** |
| | | SAM | 1.788 | 1.296 | 1.251 | 1.198 | **1.126** | 1.188 | 1.159 |
| | 4 | PSNR | 31.183 | 32.188 | 33.684 | 33.701 | 33.715 | 33.284 | **33.730** |
| | | SSIM | 0.9425 | 0.9552 | 0.9692 | 0.9690 | 0.9692 | 0.9657 | **0.9693** |
| | | SAM | 3.404 | 2.610 | 2.512 | 2.427 | 2.383 | 2.498 | **2.368** |
| Houston | 2 | PSNR | 35.286 | 37.488 | 38.503 | 38.742 | 38.765 | 38.649 | **39.036** |
| | | SSIM | 0.9937 | 0.9964 | 0.9971 | 0.9973 | 0.9973 | 0.9972 | **0.9975** |
| | | SAM | 1.319 | 1.028 | 1.001 | 0.966 | 0.967 | 0.962 | **0.960** |
| | 4 | PSNR | 28.115 | 29.747 | 31.749 | 31.511 | 32.251 | 31.444 | **32.439** |
| | | SSIM | 0.9648 | 0.9774 | 0.9858 | 0.9847 | 0.9874 | 0.9848 | **0.9881** |
| | | SAM | 2.806 | 2.093 | 1.901 | 2.014 | **1.826** | 1.932 | 1.847 |

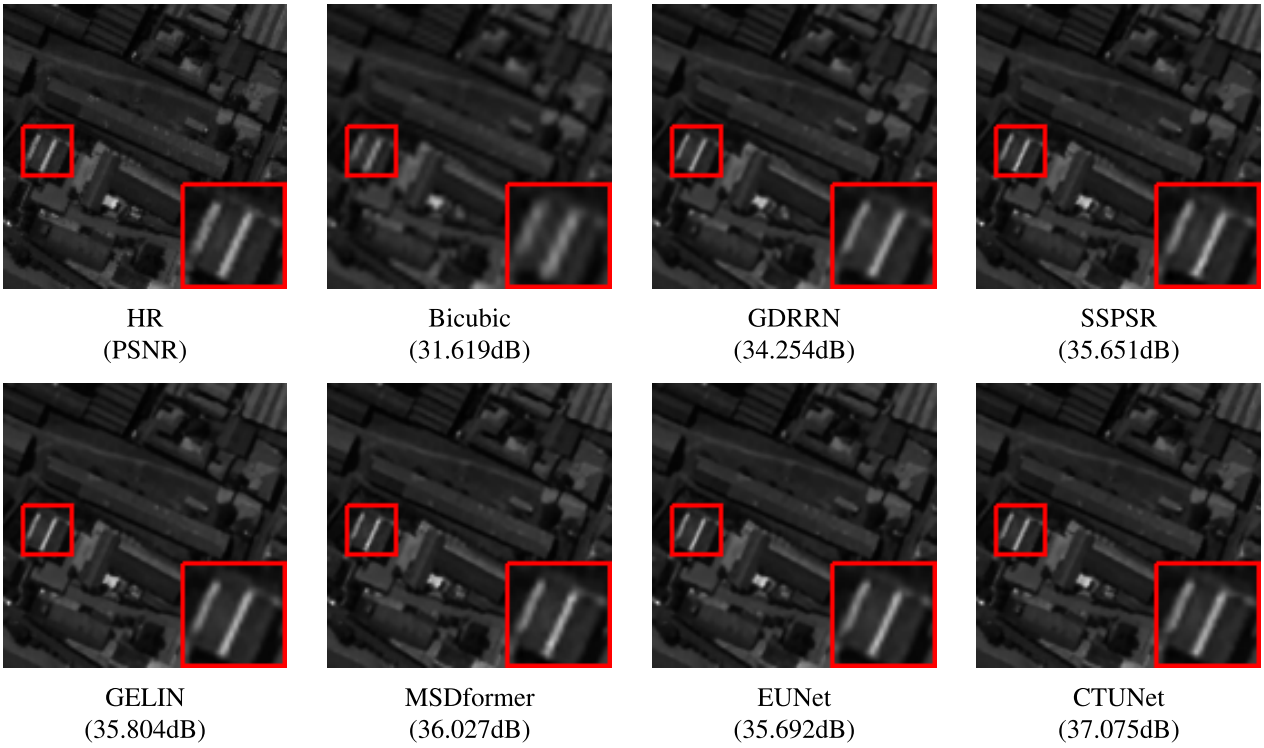



Fig. 5. Visual results of a reconstructed image in the 42nd band of Pavia Center at the scale factor ×2.

achieving the best performance in both 2-D spatial restoration and 1-D spectral consistency preservation. This relies on its integration of knowledge-driven and deep learning approaches, fully utilizing the local information extraction capability of CNNs and the global information perception capability of transformers.

From a numerical standpoint, the proposed CTUNet achieves the best results in both two levels of SR tasks when the number of bands is either low or high. On the Pavia Center dataset, our method outperforms the second-best EUNet by approximately 1 dB in terms of PSNR, demonstrating not only that unfolding networks can handle situations with limited

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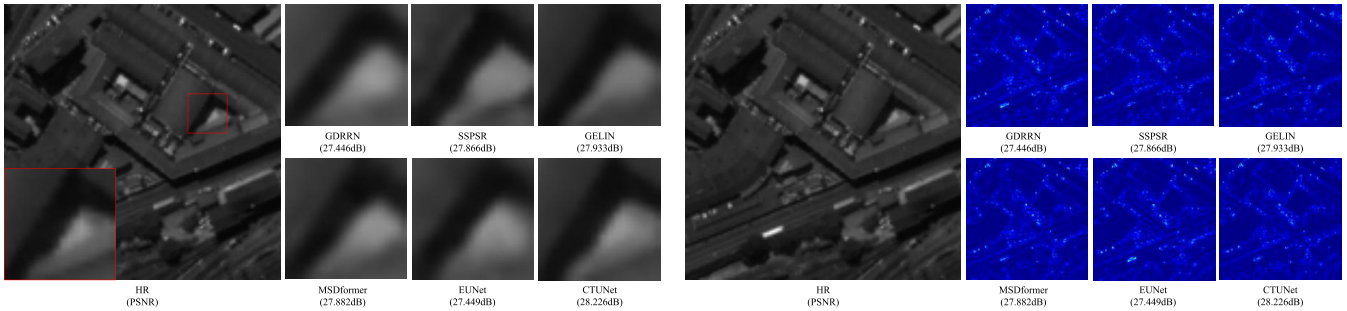



Fig. 6. (Left) Visual results of a reconstructed image in the 42nd band of Pavia Center at the scale factor $\times 4$. (Right) Visual results of the error map of a selected image of Pavia Center at the scale factor $\times 4$.

training samples effectively but also proving the accuracy of the images reconstructed by our method. Moreover, for the Houston dataset, which contains more complex land covers, our method surpasses others by more than 0.3 dB in both levels, with SAM consistently having the lowest values. These quantitative results collectively demonstrate the excellent performance of the proposed method on the datasets from different sources.

*2) Visual Results:* We selected some reconstructed images, error images, and spectral curves to compare different methods and support quantitative results, as shown in Figs. 5–11.

Fig. 5 shows the reconstructed images of a sub-image from the Pavia Center dataset in the 42nd band when scale factor is 2 ($s = 2$), with the same region enlarged by the red rectangle for better visualization of the details. The visual results indicate that CTUNet produces the most visually satisfying results, i.e., it recovers more precise and clear edge features. This could be attributed to the global learning capability, which adequately learns similar texture structural features throughout the entire image, leading to great reconstruction performance in building edges and textures. To better demonstrate the spatial reconstruction performance, we further enlarge an area of a reconstructed sub-image and visualize the error images of the sub-image when $s = 4$ in Fig. 6, which intuitively displays the SR reconstruction accuracy of each pixel on the test image, and our method provides the fewest red pixels and reconstruction errors. Additionally, from a spectral perspective, we present the spectral curves when $s = 2$ and $s = 4$ in Fig. 7, where the proposed method fits most closely with the ground-truth image, indicating that CTUNet better preserves spectral features compared to other methods.

We further demonstrate the reconstructed images from the Chikusei dataset captured by different sensors to validate the excellent performance of the proposed method. In Fig. 8, the reconstruction image at $s = 2$ is displayed, focusing on the field ridge area with the richest texture details. CTUNet is capable of providing SR images that are closest to the ground-truth image and have the clearest edge reconstruction, indicating that the proposed method can better discern and recover high-frequency information. Fig. 9 shows the reconstructed details and the SAM error map at $s = 4$, which demonstrate pixel-level spatial recovery and spectral consistency. The results show that the proposed method achieves the best spatial–spectral fidelity, while other competing methods result in more missing details and spectral distortions.

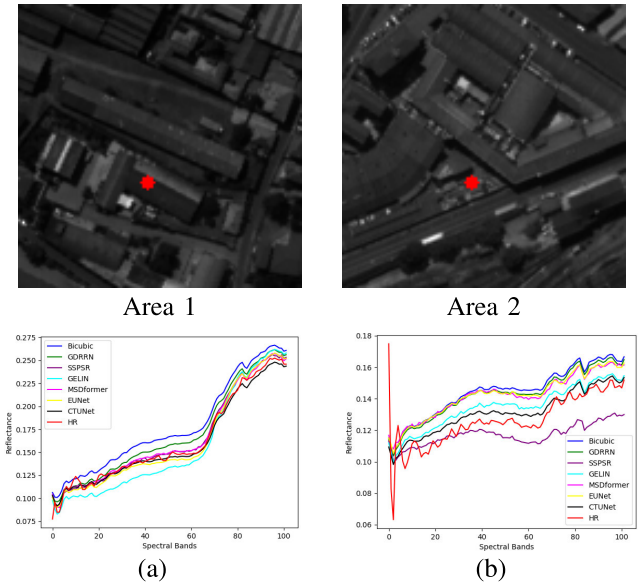



Fig. 7. Example spectral curves on two selected test areas of Pavia Center. (a) Spectrum of the red point in Area 1. (b) Spectrum of the red point in Area 2.

The Houston dataset comprises only 48 bands, and we also present its reconstructed results at level $s = 2$. As shown in Fig. 10, the proposed network demonstrates superior capability in recovering some detailed lines. From the spectral aspect, Fig. 11 displays the spectral curves of the Houston dataset at pixel $(73, 53)$, where its simple spectral pattern enables our method to approach the ground-truth image more closely in terms of spectral indices. This further demonstrates the advantage of our method in leveraging spectral similarity.

*3) Real Experiments:* We further conducted real experiments on the Pavia University dataset, directly inputting the original images into the $\times 2$ SR model trained on the Pavia Center dataset. Note that its last band is removed to ensure the same number of bands as the Pavia Center dataset. The visualized SR results are shown in Fig. 12, which provides reconstructed composite images with R–G–B in the 28–13–5 bands. It is obvious that our method restores the most subtle and natural image details. Considering the absence of ground-truth images, we adopted the blind/referenceless image spatial quality evaluator (BRISQUE) to evaluate the quality of each

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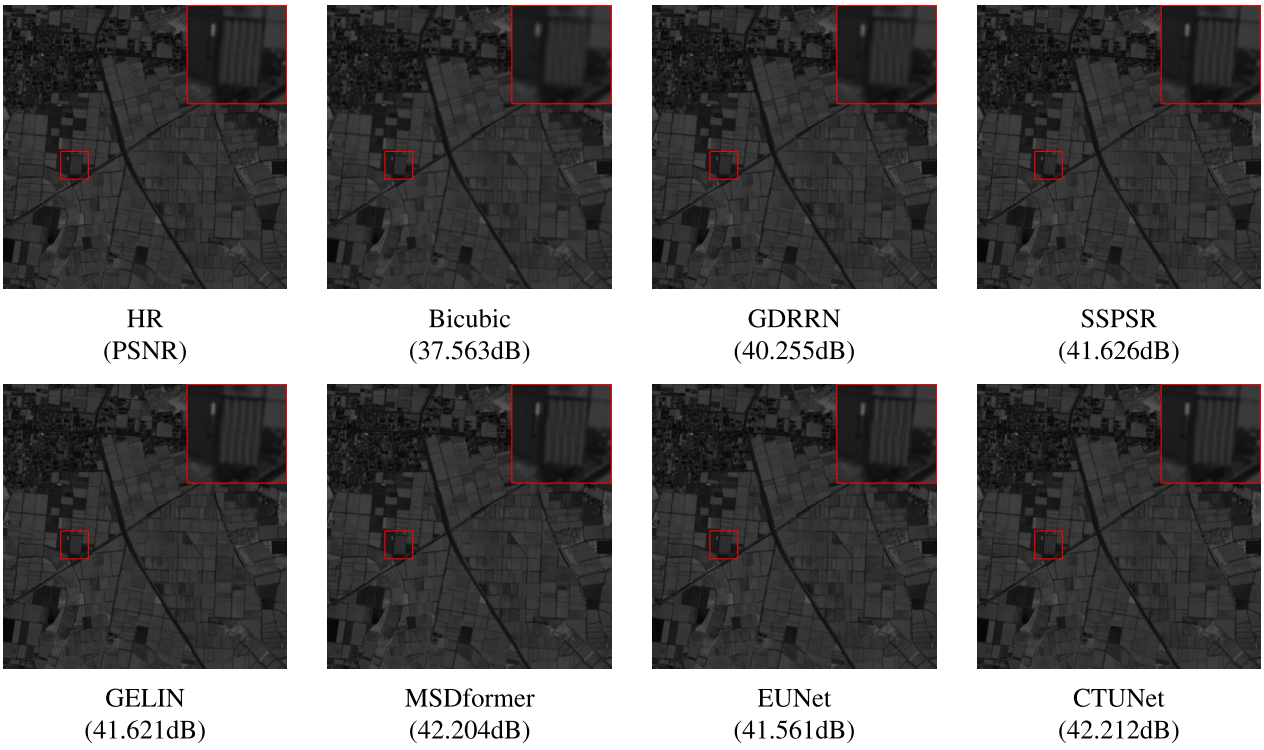



Fig. 8. Visual results of a reconstructed image in the 85th band of Chikusei at the scale factor $\times 2$.

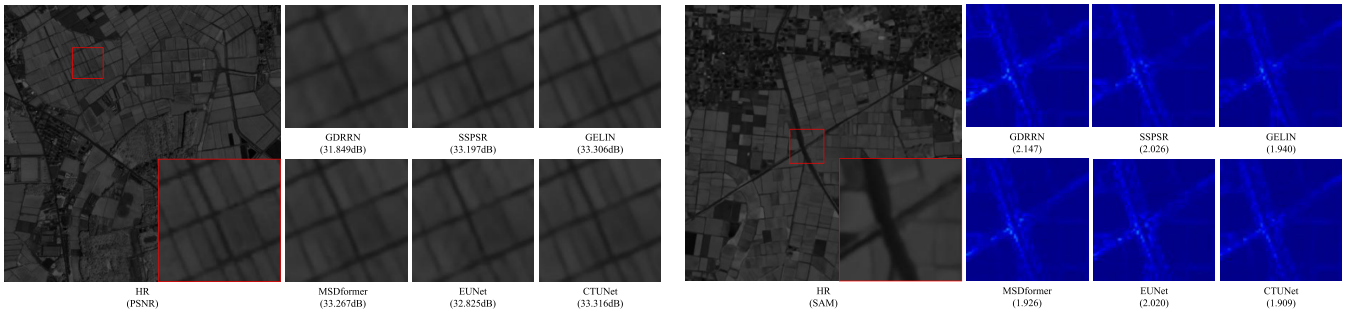



Fig. 9. (Left) Visual results of a reconstructed image in the 90th band of Chikusei at the scale factor $\times 4$. (Right) Visual results of spectral consistency on a selected image of Chikusei at the scale factor $\times 4$.

SR image [61]. The smaller the score, the better the perceptual quality, while the corresponding BRISQUE value is displayed below the SR image, where our method provides the best results. The results of real experiments confirm that CTUNet exhibits more robust and accurate SR performance in real scenarios.

### C. Ablation Study and Discussion

In this section, we conducted experiments on the Pavia Center dataset when the scale factor is $\times 2$ to discuss the effectiveness of modules and parameter settings in network design.

*1) Effectiveness of Modules:* Table II presents the ablation study of the core modules, including MSSC, EGSA, WAB, and ISIF. For the ablation experiments of the first three modules, we simply removed the corresponding modules while keeping the overall framework unchanged. When the MSSC module was removed, the PSNR of the reconstructed image decreased by approximately 1.1 dB, indicating that the multiscale information in MSSC can effectively restore each pixel in space. As essential components of the EGTB, EGSA and WAB jointly learn rich spectral information from the LR image, which has a significant impact on restoring image texture information and preserving spectral consistency. "CTUNet w/o ISIF" denotes the removal of the structure of inter-stage deep feature connections, replacing the deep features of the current stage with those of the previous stage only. The experimental results show that replacing this module leads to significant degradation in both spatial and spectral metrics, demonstrating that the information interaction across stages learns effective spatial and spectral representations.

*2) Parameter Analysis:* The proposed network has two crucial parameters: the LGFRB number $K$ and the number of iterations $T$. As shown in Table III, we not only present specific quantitative results but also report the number of parameters. By constructing different combinations of depths and studying the impact of $K$ and $T$ on evaluation metrics,

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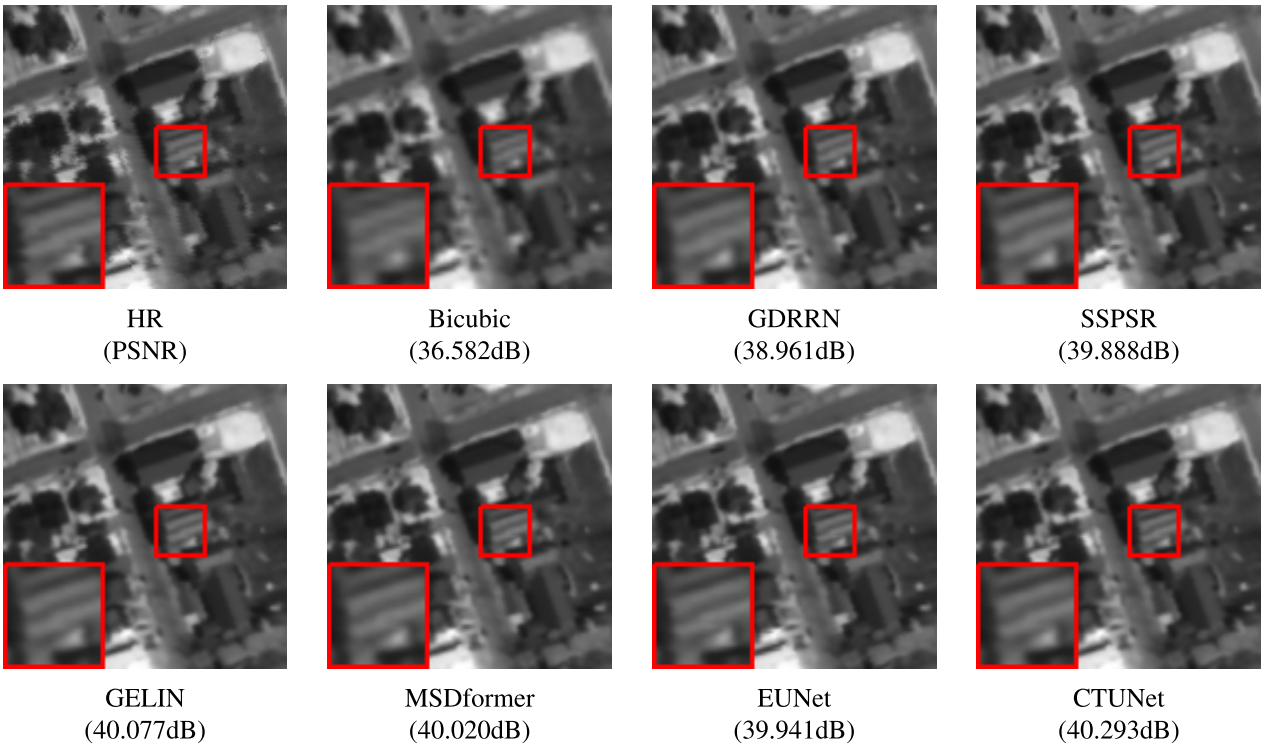



Fig. 10. Visual results of a reconstructed image in the 35th band of Houston at the scale factor $\times 2$.

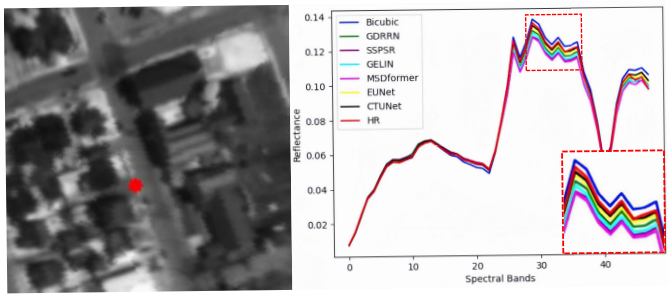



Fig. 11. Example spectral curves on a selected test areas of Houston.

TABLE II

ABLATION INVESTIGATION OF THE PROPOSED MODULES FOR SCALE FACTOR $\times 2$ ON THE PAVIA CENTER DATASET

| Architecture | PSNR | SSIM | SAM |
|---|---|---|---|
| CTUNet w/o MSSC | 34.869 | 0.9807 | 3.056 |
| CTUNet w/o EGSA | 35.658 | 0.9833 | 2.942 |
| CTUNet w/o WAB | 35.694 | 0.9834 | 2.957 |
| CTUNet w/o ISIF | 34.368 | 0.9786 | 3.250 |
| CTUNet | **35.970** | **0.9842** | **2.906** |

TABLE III

PARAMETER ANALYSIS OF CTUNET WITH DIFFERENT VALUES OF $K$ AND $T$

| Parameter Values | PSNR | SSIM | SAM | params |
|---|---|---|---|---|
| K=1 / T=1 | 35.384 | 0.9825 | 2.994 | 1548850 |
| K=2 / T=1 | 35.564 | 0.9831 | 2.946 | 2217275 |
| K=3 / T=1 | 35.826 | 0.9838 | 2.916 | 2885700 |
| K=4 / T=1 | 34.166 | 0.9777 | 2.868 | 3554125 |
| K=3 / T=2 | 35.864 | 0.9840 | 2.908 | 2885895 |
| K=3 / T=4 (Ours) | **35.970** | **0.9842** | **2.906** | 2886285 |
| K=3 / T=6 | 35.969 | 0.9841 | 2.918 | 2886675 |

we select the most suitable parameters to determine the network. Initially, the iteration count is fixed, and as more LGFRBs are stacked, the parameter number increases linearly. The network performance first slowly increases due to the increasing number of feature extraction modules. However, after reaching an LGFRB number of 3, it rapidly declines, possibly due to overfitting caused by limited training samples. Considering computational burden, we choose to fix the number of LGFRBs at 3. Subsequently, we discuss the value of $T$ after fixing $K$. When $T$ is set to 4, the model's performance reaches its peak, followed by a slight decrease, indicating that our network could achieve satisfactory results with a minimal number of iterations. Based on the above experimental results, the final network model is selected with $K = 3$ and $T = 4$.

*3) Discussion About MSSC Structure:* SCConv [51] adaptively constructs random spatial and spectral dependencies around each spatial position in the HSI, thereby enhancing the diversity of output features. Therefore, we replace the MSSC module with SCConv for comparison. Additionally, we remove the upsampling and downsampling operations in the top branch of MSSC. According to Table IV, our design achieves an increase of approximately 0.26 dB in PSNR compared to SCConv, and there is also a significant improvement in spectral metrics. The results of "MSSC w/o Scale" also demonstrate that the operation of increasing image

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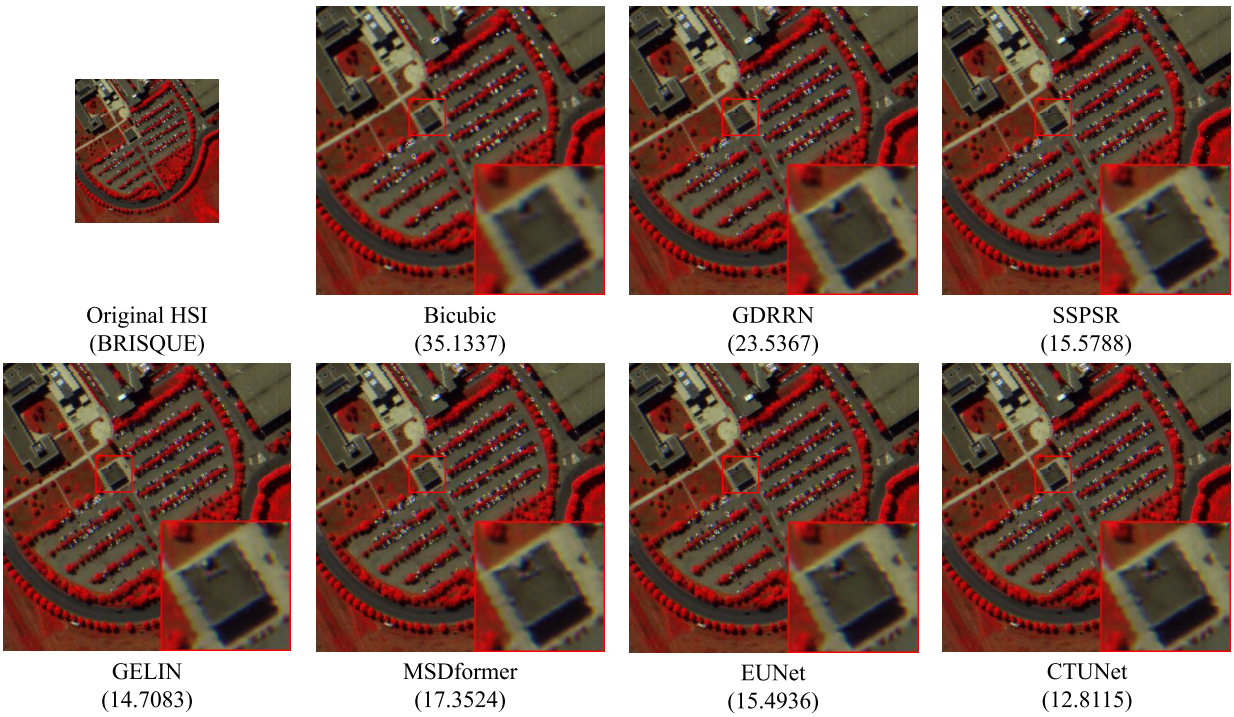



Fig. 12. Real experiment results on Pavia University.

TABLE IV
DISCUSSION ABOUT MSSC STRUCTURE FOR SCALE FACTOR ×2 ON THE PAVIA CENTER DATASET

| Architecture | PSNR | SSIM | SAM |
|---|---|---|---|
| SCConv | 35.705 | 0.9834 | 2.934 |
| CTUNet w/o Scale | 35.932 | 0.9837 | 2.929 |
| CTUNet | **35.970** | **0.9842** | **2.906** |

TABLE V
DISCUSSION ABOUT EGTB STRUCTURE FOR SCALE FACTOR ×2 ON THE PAVIA CENTER DATASET

| Value | PSNR | SSIM | SAM |
|---|---|---|---|
| $\alpha = 0$ | 35.932 | 0.9840 | 2.928 |
| $\alpha = 0.01$ (Ours) | **35.970** | **0.9842** | **2.906** |
| $\alpha = 0.1$ | 35.855 | 0.9839 | 2.944 |
| $\alpha = 1$ | 35.854 | 0.9839 | 2.912 |

TABLE VI
DISCUSSION ABOUT WAB STRUCTURE FOR SCALE FACTOR ×2 ON THE PAVIA CENTER DATASET

| Architecture | PSNR | SSIM | SAM |
|---|---|---|---|
| WAB w/o BigConv | 35.765 | 0.9837 | 2.915 |
| WAB w/o DilatedConv | 35.856 | 0.9839 | 2.937 |
| WAB | **35.970** | **0.9842** | **2.906** |

scale allows the network to perceive a wider range of image features, further enriching the spatial–spectral representation.

*4) Discussion About EGTB Structure:* We explore the proportion $\alpha$ of the EBGM output features in this module, setting it to 0, 0.01, 0.1, and 1, respectively. Experimental results shown in Table V demonstrate that injecting a small amount of edge information into the non-local module can effectively improve the learning of spatial–spectral features in the image. However, if there is too much edge information, it may lead to a certain degree of performance degradation in the network. We ultimately determine $\alpha = 0.01$.

TABLE VII
LOSS FUNCTION ANALYSIS FOR SCALE FACTOR ×2 ON THE PAVIA CENTER DATASET

| Weight | PSNR | SSIM | SAM |
|---|---|---|---|
| $\lambda_s = 0, \lambda_g = 0$ | 35.844 | 0.9837 | 2.961 |
| $\lambda_s = 0.1, \lambda_g = 0$ | 35.689 | 0.9833 | 2.946 |
| $\lambda_s = 0.1, \lambda_g = 0.01$ | 35.970 | 0.9842 | **2.906** |
| $\lambda_s = 0.1, \lambda_g = 0.1$ | 35.872 | 0.9840 | 2.925 |
| $\lambda_s = 0.01, \lambda_g = 0.01$ | **35.993** | **0.9843** | 2.923 |

*5) Discussion About WAB Structure:* To investigate the role of the "large receptive field" in WAB, we replace the original dilated convolutions with ordinary 3 × 3 convolutions (WAB w/o BigConv) or did not use dilated convolutions (WAB w/o DilatedConv), resulting in smaller receptive fields. The results shown in Table VI demonstrate that a large receptive field can greatly improve the feature extraction ability, especially in spatial dimension reconstruction.

*6) Discussion About Loss Function:* The weight of MAE loss $\lambda_s$ and the weight of gradient loss $\lambda_g$ are set to compose different loss functions, which are employed to train the proposed CTUNet, respectively. The quantitative results are shown in Table VII. Obviously, the constraint of SAM loss

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TABLE VIII
MODEL COMPLEXITY COMPARISON

| Network | GDRRN | SSPSR | GELIN | MSDformer | EUNet | CTUNet |
|---|---|---|---|---|---|---|
| PSNR(dB) | 33.478 | 34.853 | 34.820 | 35.138 | 34.898 | 35.970 |
| Params | 382464 | 11024334 | 13116928 | 12833390 | 714241 | 2886285 |
| Inference time($10^{-2}$s) | 0.898 | 9.076 | 6.582 | 7.879 | 1.744 | 5.558 |

can effectively reduce the value of SAM, but at the cost of a slight decrease in PSNR. On the other hand, the imposition of gradient loss with small weight can enhance the PSNR and SAM values, avoiding blurry textures and edges. To achieve a balance between spatial and spectral metrics, after comparing different weight combinations, we fix $\lambda_s = 0.1$ and $\lambda_g = 0.01$.

*7) Model Complexity:* We show the reconstruction performance (measured by PSNR) versus the number of network parameters and the inference time of six network models at the scale factor of $\times 2$ on the Pavia Center dataset in Table VIII. The early method GDRRN, with its simple network structure, unsurprisingly has the fewest parameters and shortest inference time, but this also means its reconstruction performance is the lowest. Subsequently, recent methods such as SSPSR, GELIN, and MSDformer maintain a large number of parameters during training, placing a relatively heavy burden on the computational resources, causing an increase in inference time. The use of the unfolding framework allows us to drastically lighten the network structure at each iteration. Therefore, EUNet achieves satisfactory reconstruction performance while ensuring fewer parameters. Our proposed CTUNet also utilizes the unfolding structure and explores the design of the network structure. In comparison, CTUNet achieves the highest quality while maintaining competitive parameter numbers and inference time.

## V. CONCLUSION

In this article, we propose a novel CTUNet for HSI SR. The proposed structure considers the degradation process of HSIs, employing an iterative optimization strategy and embedding deep learning networks within it. To capture multiscale information in HSIs, we design MSSCs for local and regional feature learning. Considering the spectral correlation of HSIs, we improve the traditional vision transformer by designing an EGTB for global feature learning, while using large kernel dilated convolutions in the feedforward part to enlarge the receptive field. Additionally, connections between adjacent iterations are established to model and learn implicit spatial–spectral priors. Comprehensive ablation experiments validate the effectiveness of each module in spatial resolution enhancement and spectral consistency. Comparative experiments conducted on three publicly available datasets confirm that our method achieves higher reconstruction performance and visual quality compared to several state-of-the-art HSI SR methods.

In the future work, we will investigate more novel and more efficient learning structures, such as diffusion models, to enhance the network's robustness and adaptability, thereby handling a broader range of datasets from different sources. In addition, we will attempt to explore the degradation factors that occur during the imaging process and propose more targeted refinement stages to enhance the effectiveness and practical applications of the SR task.

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**Yao Tang** received the B.S. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2022, where he is currently pursuing the M.S. degree with the School of Geodesy and Geomatics.

His research interests include hyperspectral image super-resolution, quality improvement, deep learning, and computer vision.

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**Jie Li** (Member, IEEE) received the B.S. degree in sciences and techniques of remote sensing and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2011 and 2016, respectively.

He is currently an Associate Professor with the School of Geodesy and Geomatics, Wuhan University. His research interests include image quality improvement, image super-resolution reconstruction, data fusion, remote sensing image processing, sparse representation, and deep learning.



**Linwei Yue** (Member, IEEE) received the B.S. degree in geographic information system and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2012 and 2017, respectively.

She is currently an Associate Professor with the School of Geography and Information Engineering, China University of Geosciences, Wuhan. Her research interests include multi-source remote sensing data fusion and water environment monitoring.



**Xinxin Liu** (Member, IEEE) received the B.S. degree in geographic information system and the Ph.D. degree in cartography and geographic information system from Wuhan University, Wuhan, China, in 2013 and 2018, respectively.

In July 2018, she joined the College of Electrical and Information Engineering, Hunan University, Changsha, China, where she is currently an Associate Professor. Her research interests include image quality improvement, remote sensing image processing, and remote sensing mapping and application.



**Yajie Li** received the B.S. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2021, where she is currently pursuing the M.S. degree with the School of Geodesy and Geomatics.

Her research interests include hyperspectral denoising, quality improvement, remote sensing image processing, and deep learning.



**Yi Xiao** (Graduate Student Member, IEEE) received the B.S. degree from the School of Mathematics and Physics, China University of Geosciences, Wuhan, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Geodesy and Geomatics, Wuhan University, Wuhan.

His major research interests include remote sensing image/video processing and computer vision. More details can be found at https://xy-boy.github.io/.



**Qiangqiang Yuan** (Member, IEEE) received the B.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2006 and 2012, respectively.

In 2012, he joined the School of Geodesy and Geomatics, Wuhan University, where he is currently a Professor. He published more than 90 research papers, including more than 70 peer-reviewed articles in international journals such as IEEE TRANSACTIONS ON IMAGE PROCESSING and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include image reconstruction, remote sensing image processing and application, and data fusion.

Dr. Yuan was a recipient of the Youth Talent Support Program of China in 2019 and the Top-Ten Academic Star of Wuhan University in 2011. In 2014, he received Hong Kong Scholar Award from the Society of Hong Kong Scholars and China National Postdoctoral Council. He is on the editor board of nine international journals, and has frequently served as a referee for more than 50 international journals for remote sensing and image processing.

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